National University of Computer and Emerging Sciences

Database Management System Lab

Project Deliverable 1: ERD Design

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SECTION:	B
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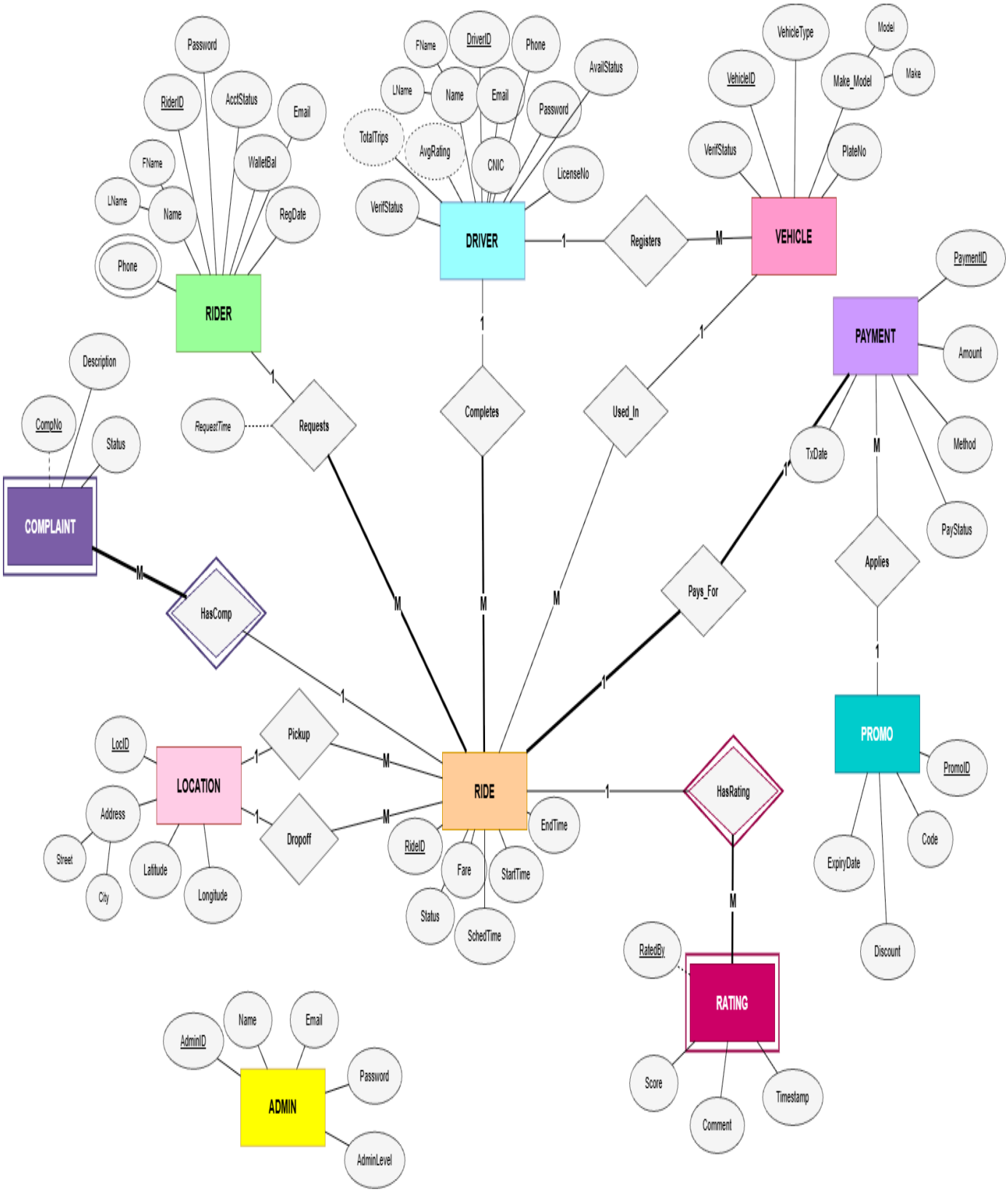
1. ERD Design


RIDER
RiderID
Password
AcctStatus
Email
WalletBal
RegDate
FName
LName
Name
Phone
DRIVER
DriverID
FName
LName
Name
Email
Phone
Password
AvailStatus
TotalTrips
AvgRating
CNIC
LicenseNo
VerifStatus
Registers
VEHICLE
VehicleID
VehicleType
Model
Make_Model
Make
VerifStatus
PlateNo
PAYMENT
PaymentID
Amount
Method
PayStatus
TxDate
COMPLAINT
CompNo
Description
Status
Requests
RequestTime
Completes
Used_In
Pays_For
Applies
HasComp
LOCATION
LocID
Address
Street
City
Latitude
Longitude
Pickup
Dropoff
RIDE
RideID
Fare
Status
SchedTime
StartTime
EndTime
HasRating
PROMO
PromoID
Code
ExpiryDate
Discount
RATING
RatedBy
Score
Comment
Timestamp
ADMIN
AdminID
Name
Email
Password
AdminLevel

2. Entities and Attributes

Based on the RideFlow functional modules, the following entities are defined:

- **Users:** The central repository for all accounts. Includes UserID (PK), FirstName, LastName, Email, Phone, PassHash, Role (Admin/Rider/Driver), Status (Active/Banned), and RegDate.
- **Driver:** A specialized entity linked to a User account. Includes DriverID (PK), LicenseNo, CNIC, ProfilePhoto, VerifStatus, AvailStatus, and WalletBalance.
- **Vehicle:** Records for transport units. Includes VehicleID (PK), Make, Model, LicPlate, Color, Type (Economy/Premium/Bike), and VerifStatus.
- **Ride:** The core transactional entity. Includes RideID (PK), Fare, SurgeMultiplier, RideStatus (Requested/Accepted/In-Progress/etc.), SchedTime, StartTime, and EndTime.
- **Location:** Stores spatial data for efficiency. Includes LocationID (PK), Latitude, Longitude, Address, and City.
- **Payment:** Financial records. Includes PaymentID (PK), Amount, Date, PayStatus, and PayMethod (Cash/Wallet/Card) .
- **Rating (Weak):** Post-trip feedback. Attributes include Score (1-5), Comment, and Timestamp.
- **Complaint (Weak):** Customer service logs. Attributes include Subject, Description, and Status.

3. Key Relationships and Cardinalities

- **Users to Driver (1:1):** A user account may optionally act as a driver profile. The Driver profile cannot exist without a parent User .
- **Users to Ride (1:N):** A user (rider) can request multiple rides over time, but each ride is initiated by exactly one user .
- **Driver to Ride (1:N):** One driver can complete many rides, but each active ride must have one assigned driver .
- **Driver to Vehicle (1: N):** A driver can register multiple vehicles (e.g., a bike and a car), but each vehicle is owned by one driver.
- **Ride to Payment (1:1):** Every completed ride generates exactly one payment record.
- **Ride to Location (N:1):** Each ride is linked to two locations: one for Pickup and one for Drop-off.

4. Advanced Database Logic

To satisfy the full project scope, the following logic will be implemented in the database:

- **Stored Procedure:** Will be developed to calculate fares using the formula: $Base Rate + (Per KM \times Distance) + (Per Minute \times Duration)$, then applying the Surge Multiplier.
- **Trigger:** A trigger will monitor the Rating table to automatically flag any Driver whose average score falls below 3.5 stars.
- **View:** A Leaderboard view will be created to display top-rated drivers by city.
- **DCL:** Role-based access control will be managed via GRANT and REVOKE to ensure Riders cannot access Admin reports.